Ashmita Dua

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Education

University of California, Santa Cruz | Undergraduate **Sept, 2024 - June, 2028**

Bachelor of Science (B.Sc.) in Computer Science | GPA 3.71/4.0 (Dean's Honor List)

Relevant Coursework- CSE101 (Data structures/algorithms), CMPM80K (Game Engines), CSE40 (AI/ML)

Skills / Tech Stacks

Game Development: Unity/C#, Procedural Content Generation (WFC), Level Design, Godot

Programming Languages: Python, C/C++, JavaScript/TypeScript, HTML/CSS,

Tools & Frameworks: Git, LangChain/LangGraph, OpenCV, APIs, Cursor, Kubernetes, Claude Code, Multi-Agent Envs

AI/ML: LLMs (Gemini, Claude), RAG, PyTorch, CNNs, Reinforcement Learning, Fine-tuning, NeuroSymbolic AI

Experience

Kaido: Trainer — **April, 2026 - Present**

LLM Integration Specialist

- Designing LLM-powered companion AI for an educational electronics RPG on a portable handheld device
- Architecting conversational interface integrating LLM dialogue generation with circuit-building gameplay progression
- Leading AI/game integration as sole LLM specialist on a 3-person pre-launch startup team

Terraforma by Overture — **March, 2026 - Present**

Open-Source Contributor

- Building an agentic AI pipeline that corrects stale location data in the Overture Maps Foundation Places dataset using LLM-powered web cross-referencing
- Developing confidence scoring system to flag closures, moved businesses, and outdated entries across geographic regions
- Engineering full-stack review dashboard (React + FastAPI + PostgreSQL) with interactive map view for human-in-the-loop correction approval
- Designing neurosymbolic layer to complement LLM agent reasoning with structured logical rules for data quality verification

Augmented Design Lab — **September, 2025 - Present**

Undergraduate Researcher

- **Reduced LLM validation loop errors by 42%** (78.7% → 46.3%) in the level generation system by redesigning system prompts to enforce action-first behavior and structured tool use, in Pewter Platformer (See Projects section)
- Led a study focused on generating game levels through LLMs/PCG and hierarchical wave function collapse
- Implemented action-first behavioral constraints, reducing repetitive agent loops across **30+ test scenarios**

Artificial Intelligence Explainability Accountability Lab — **April, 2025 - Present**

Undergraduate Researcher

- Leading a study focused on Episodic Memory using Neuro Symbolic layers, and query agnostic KV
- Configured local LLM API Memory Management and retrieval models and integrated SWI-Prolog for logical reasoning verification

Undergraduate Admissions Office, UCSC — **December, 2024 - Present**

Website Designer

- Maintain university admissions website serving **50,000+ prospective students** annually
- **Redesigned 15+ web pages** improving mobile responsiveness and navigation
- Managed structured data systems supporting admissions cycle workflows

Sowiz — **June, 2025 - July, 2025**

AI/ML Intern

- Engineered **RAG workflows using LangChain** to create an intelligent agent system for dealer support and product knowledge
- Designed and implemented gamified sales platform with a milestone-based reward system to incentivize unit sales performance
- Integrated AI-powered information retrieval system enabling dealers to access real-time product specs and pricing
- Improved dealer conversion rates by providing comprehensive, accessible resources through an automated workflow system

Publications

Pewter: A Mixed-Initiative System for Natural Language Tilemap Design

Under review at UIST '26 Conference | Associated with Augmented Design Lab (w. Prof. Jim Whitehead)

A mixed-initiative system enabling natural language-driven tilemap level design, combining LLM agents with hierarchical wave function collapse to reduce validation loop errors by 42% across 30+ test scenarios.

Engram: Personality-Parameterized Schema Memory for NPC Cognitive Diversity

Under review at FDG '26 | Co-authored w. Devesh Kriplani | Demo: https://bit.ly/engramPersonalityNPC

Parameterizes OCEAN (Big Five) personality traits with Prolog-based memory encoding rules, enabling NPCs to form structurally different memories from identical experiences and producing trait-consistent cognitive diversity without scripting.

Projects

CopCuts, Barber LLM-based NPCs Unity Game (Portfolio Project)

- Building a multi-agent LLM-powered NPC detective game, trained on Reinforcement Learning, using 3D game logic.
- LLM-powered featuring dynamic conversations, memory systems, and personality adaptation through prompt engineering
- Implemented suspicion mechanics and relationship tracking where NPC responses vary based on player interrogation strategy

Woven Mood-Based Storytelling LLM | Demo: duashmita.github.io/woven/ | Repo: bit.ly/woven-intoyou

Presented at the Undergraduate Student Research Showcasing 2025 with AIEA under Prof. Leilani Gilpin

- Designed LLM-powered narrative system guiding users from current emotion to target mood across 8-10 turn interactive stories
- **Achieved 87% emotional arc accuracy** by implementing rule-based validation with a 10-emotion NLP classifier
- Compared GPT-4 vs Gemini performance: Gemini averaged within 2 mood-indices of target; GPT-4 remained 1-3 indices off

HabPet | Demo: bit.ly/HabpetYTDemo | Web: habit-bot.vercel.app | Repo: github.com/Sanyab8/habit-bot

Project for CruzHacks'26

- Built an IoT habit tracker combining ESP32 hardware with a web app featuring AI-powered habit detection via OpenCV and progressive milestone rewards
- Engineered Node.js bridge for real-time ESP32-web communication using sensors and actuators (servos, LEDs, buzzers)
- Deployed full-stack application using React/Vite/Zustand on the Vercel platform

Pewter Platformer | Demo: bit.ly/3NVYIYt

Associated with Augmented Design Lab (w. Prof. Jim Whitehead)

- Diagnosed an LLM failure mode in **TypeScript,** where the agent entered repetitive validation loops during level creation
- Iterated the system prompt + interaction flow to enforce 'action-first' behavior and structured tool use
- Benchmarked over 30 test prompts: loop-rate **78.7% → 46.3%**

Certificates

Goldman Sachs Possibilities Summit 2025 | Selected Delegate | Jan 2025

1 of 100 students selected nationally for leadership and career development program